Energy Conversion and Management 244 (2021) 114521

Contents lists available at ScienceDirect

Energy Conversion and Management

journal homepage: www.elsevier.com/locate/enconman

Review

Efficiency improvement of passively cooled micro-scale hybrid CPV-TEG systems at ultra-high concentration levels

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ARTICLE INFO

Keywords:
Concentrator photovoltaics
Efficiency improvement
Finite-element analysis
Hybrid system
Thermoelectric generator
Ultra-high concentration

ABSTRACT

Concentrator photovoltaic technology faces the challenge of evolving toward ultra-high concentration levels (>2,000 suns) to increase the conversion efficiency of sunlight to electricity and lower the system cost. Recent studies have analysed technical issues related to optics, concentrator solar cell development, and thermal management. On the contrary, the hybridisation of thermoelectric generators with concentrator systems is a promising way to recover part of the waste heat generated in the solar cells and convert it to electricity, thereby enhancing global efficiency. There is theoretical evidence that increasing the concentration factor favours this type of hybridisation. However, the investigation of hybrid thermoelectric concentrator photovoltaic systems at ultra-high concentration levels is pending. In this study, a detailed 3D finite-element approach, which is also validated with previous literature, is used to analyse the efficiency improvement achievable with these hybrid systems at concentration factors of up to 4,000 suns using passive cooling. The use of micro-scale solar cells reduces the thermal requirements, making the use of passive cooling mechanisms feasible. The influence of a number of design parameters (such as solar cell area, temperature coefficient of efficiency, area ratio, thermoelectric leg height, fill factor, and ZT figure-of-merit or heat-exchanger thermal resistance) in the system thermal and electrical performance was assessed. With this information, a future optimum hybrid solar receiver is numerically evaluated assuming that several technological constraints present in the state-of-the-art concentrator receivers are overcome, i.e. that microcells of 0.5 × 0.5 mm^2 at 4,000 suns operating at 150 °C with extremely low-temperature coefficient of efficiency and very low heat-exchanger thermal resistance can be manufactured in next years. The results demonstrate that the use of thermoelectric generators with a high ZT could improve the efficiency of a concentrator photovoltaic-only receiver using the same solar cell at the same concentration up to 10.8% in relative magnitude. This study should encourage researchers to solve the problems associated with ultra-high concentrations of micro-scale high-temperature multi-junction solar cells integrated with high-efficiency optics, high ZT thermoelectric materials, and powerful cooling mechanisms.

1. Introduction

Concentrator photovoltaic (CPV) systems are a promising technology to increase the efficiency of sunlight conversion to electricity and reduce the cost of energy production. A CPV system is typically composed of an array of electrically interconnected solar cells mounted on receivers, optical elements capable of efficiently concentrating direct sunlight onto the solar cells with sufficient angular tolerance, a cooling system to remove heat from the solar cell, mechanical subsystems such as protecting elements against environmental conditions, and a two-axis solar tracker, keeping them perpendicular to the sun's rays at all times [1]. These systems use optical light concentrators to increase the incident power of the solar cells. Depending on the concentration level, the CPV systems can be classified as low (<10 suns), medium (10–100 suns), high (100–2,000 suns), and ultra-high (>2,000 suns) concentrations [2]. The solar cell area required in CPV is comparatively small, thus saving expensive semiconductor materials and allowing the use of more sophisticated multi-junction (MJ) solar cells. Typical solar cells employed in CPV are the MJ that have reached the highest conversion efficiencies among all terrestrial photovoltaic (PV) applications, with efficiency records of 44.4% for 3-junction, 46.0% for 4-junction, and 47.1% for 6-junction solar cells [3,4]. CPV modules can reach efficiency records of 40% at the prototype level and 30% for commercial designs [5]. Despite

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https://doi.org/10.1016/j.enconman.2021.114521
Received 25 January 2021; Accepted 8 July 2021
Available online 17 July 2021
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Nomenclature

Symbols

A_{cell}	Surface area of solar cell m^2
A_{lens}	Surface area of primary optical element m^2
A_{TEG}	Surface area of thermoelectric module m^2
AR	Area ratio between thermoelectric module and solar cell -
C_g	Geometric concentration ratio -
DNI	Direct normal irradiance W/m^2
$\vec{E}$	Electric field vector V/m
G_0	Solar power on the optical system W
h_{cool}	Convective heat transfer coefficient of the cooling system $W/(m^2K)$
I_{TEG}	Current intensity of thermoelectric module A
$\vec{J}$	Current density A/m^2
k	Thermal conductivity of material W/(mK)
P_{CPV}	Electric power generated in solar cell W
P_{TEG}	Electric power generated in thermoelectric module W
$P_{CPV-TEG}$	Electric power generated in hybrid system W
Q	Conduction heat rate W
$\vec{q}$	Conduction heat rate vector W/m^2
$\dot{q}$	Heat source term of heat diffusion equation W/m^3
Ra	Rayleigh dimensionless number -
R_{cool}	Absolute thermal resistance of cooling system K/W
REI	Relative efficiency improvement %
R_L	Matching load resistance of thermoelectric module Ω
S	Seebeck coefficient V/K
T	Scalar temperature field K
T_{air}	Temperature of the air surrounding K
T_{cell}	Average temperature of solar cell K
T_{TE}	Average temperature of thermoelectric material K
$TEG_{effectiveness}$	Effectiveness of thermoelectric generator %
TC	Temperature coefficient of efficiency %/K
V	Volume of solar cell m^3
V_{TEG}	Voltage on thermoelectric module terminals V
ZT	Dimensionless figure-of-merit -

Greek letters

η_{cell}	Efficiency of solar cell %
η_{CPV}	Efficiency of concentrator photovoltaic system %
$\eta_{ref,cell}$	Reference efficiency of solar cell %
η_{opt}	Efficiency of the optical elements %
η_{TEG}	Efficiency of thermoelectric generator system %
$\eta_{TEG,max}$	Maximum efficiency of thermoelectric generator system %
σ	Electrical conductivity S/m

Subscripts

cold	Cold side of thermoelectric module
hot	Hot side of thermoelectric module
p	P-type thermoelectric material
n	N-type thermoelectric material
TM	Temperature-matched
TMM	Temperature-mismatched

Abbreviations

CPV	Concentrator photovoltaic
CSOC	Concentrator standard operating conditions
FEM	Finite-element method
LCOE	Levelised cost of energy
MPP	Maximum power point
SOE	Secondary optical element
TEG	Thermoelectric generator
TIM	Thermal interface material
UHCPV	Ultra-high concentrator photovoltaic

the high efficiencies, the levelised cost of energy (LCOE) and cost per installed capacity of this technology still need to be reduced to compete with non-concentrating PV technologies [6]. One promising way to lower the cost and increase the competitiveness of CPV is to develop ultra-high CPV systems (UHCPV) with concentration ratios > 2,000 suns [7]. However, the development of UHCPV systems faces technological challenges regarding solar cell architecture, optical configuration, and thermal management of solar receivers.

Recent investigations in solar cells have offered solutions to eliminate the power losses caused by the series resistance at ultra-high (UH) concentrations. New designs such as the so-called vertical-tunnel-junction (VTJ) architecture [8] or duplicated junction designs [9] have shown the potential to achieve UH levels of concentration with very small degradation of solar cell efficiency. The miniaturisation of solar cells is another promising research area for UHCPV. By reducing the cell size to the sub-millimetre range, several benefits can be potentially achieved, such as material savings, cost reduction, better thermal management, resistance loss minimisation, and enhanced efficiency [10,11,12]. As the heat input per cell was reduced, the solar cell temperature also decreased. Furthermore, as the ratio of surface area to volume increases, the heat dissipation to the ambient increases [13]. Unlike conventional CPV, which sometimes requires active cooling, micro-CPV does not have this requirement [14].

Advances in optical configuration have made it possible to achieve high concentrations. Karp *et al.* designed a compact micro-optic waveguide to concentrate sunlight up to an intensity of approximately 2,500 suns, assuming an optical efficiency>80% [15]. By combining a diverging polycarbonate (PC) lens and a converging PMMA lens, overcoming the concentration limits due to optical aberration and achieving a concentration of up to ~ 8,500 suns was possible [16]. Shanks *et al.* demonstrated the possibility of achieving concentrations within the range of 3,000–4,300 suns with an optical efficiency of 55%–75% [17]. Moreover, a one-cell prototype with a 3,015 × geometrical concentration was fabricated and characterised in a CPV solar simulator as a proof of concept [18].

As the sunlight concentration increased, the cooling mechanism became crucial in the CPV. Different strategies to implement cooling mechanisms (passive or active), innovative technologies, and materials are being investigated [19,20,21]. Royne *et al.* analysed the microchannel heat sink as the best cooling system for a high-density-packed CPV system [22]. For this system, the thermal resistance (R_{th}) of the cooling system must be$<10^{-4}$ (m^2 K)/W for concentration levels above 150 suns. Studies on liquid cooling (heat pipes, liquid immersion, jet impingement, phase-change materials, natural convection of nanofluids, among others) have reported heat transfer coefficients in the range of 1,000–6,000 $W/(m^2K)$. These cooling technologies are able to remove heat flux up to 500 W/cm^2 [23] and maintain the cell temperature below 60 °C with concentration levels up to 1,000 suns [20].

Passive cooling mechanisms have been proven to successfully handle thermal management for high concentration ratios up to 2,000 suns, and the use of a metal (generally aluminium) plate heat sink is the most common solution for CPV systems [22] with typical values of heat transfer coefficient in the range of 5–1,000 $W/(m^2 K)$. In 2001, Mudawar *et al.* established a maximum achievable value of 1,600 $W/(m^2 K)$ for the heat transfer coefficient under passive cooling conditions [24]. Gualdi *et al.* studied the limits of passive cooling in CPV systems by using a 1D analytical model of a solar cell directly mounted on a flat-plate heat sink [25]. The authors proved that solar cells with areas ranging from 2 mm

× 2 mm to 10 mm × 10 mm could maintain the cell temperature below 80 °C up to concentration ratios of 2,000 suns and 350 suns, respectively. Recently, Valera *et al.* studied by 3D numerical model the potential of different receiver substrates (fabricated from aluminium and copper) as a flat-plate heat sink using a geometric concentration of 2,000–10,000 suns. The maximum operation cell temperatures were below 80 °C for small solar cells of 1 × 1 mm^2, even up to 10,000 suns [26].

The efficiency of PV systems can be improved by hybridisation with thermoelectric generators, which allows for part of the waste heat generated in the solar cells to be transformed into electricity. Although hybridisation with non-concentrating PV systems has been studied [27,28,29], their feasibility has been questioned owing to the high-temperature coefficient of efficiency of conventional solar cells and the low-temperature gradient achievable [30,31,32]. A TEG module thermally coupled with a solar cell increases the thermal resistance of the structure, thereby increasing the operating cell temperature with respect to the PV-only system as a consequence. For the high-temperature coefficients given in conventional PV materials, the degradation of cell performance with temperature dominates over the gains provided by the TEG. Even in some cases where efficiency improvement is reported, the slight energy gains cannot justify the costs of integrating TEGs with non-concentrating PV modules. On the contrary, CPV systems are more attractive for this hybridisation owing to the small temperature coefficients of MJ solar cells and the high temperature gap between the hot and cold sides of the TEG. There is theoretical evidence that increasing the concentration factor favours hybridisation [33]. For this reason, UHCPV systems seem to be even more suitable for hybridisation, although there is a lack of studies on this topic that needs to be addressed to assure their potential.

There is wide interest in thermoelectricity (TE) owing to many advantages such as high scalability, long operating life, solid-state operation with no moving parts, no chemical reactions, zero emissions or solid waste, and free maintenance [34]. Despite the low efficiencies of TE devices, their characteristics allow their use in many applications for energy harvesting, taking advantage of the wasted heat source [35]. Currently available TE materials have a *ZT* figure-of-merit of approximately 1 or lower, indicating an efficiency of 4%–6% for a typical TE device. Bismuth telluride (Bi_2Te_3) has been the most suitable material used for industrial TE modules with average values of $ZT \approx 0.5$–0.8 for temperature ranges of 400 to 500 K. The outlook for laboratories is to develop materials with $ZT \geq 2$ to achieve an efficiency above 10% [36]. High *ZT* values (up to 2.6) were reported in a thin-film p-type Bi_2Te_3/Sb_2Te_3 superlattice at 300 K [37]. A p-type alloy 25 % Bi_2Te_3 + 75 % Sb_2Te_3 was successfully manufactured by rapid solidification and hot pressing processes, achieving a maximum ZT value of 2.38 at 773 K [38].

Numerous investigations have focused on optimising the hybrid CPV-TEG system using theoretical models. However, most of these were applied at low concentrations. A model based on non-equilibrium thermodynamics theory and law of conservation of energy providing guidelines for the optimisation of the design of CPV-TEG systems up to six suns was presented in [39]. Liao *et al.* investigated a theoretical model of a hybrid system in the concentration range of 1 to 5 suns. The influence of several important factors such as the thermal conductance between the CPV and TEG, the CPV current, the solar irradiation, the concentrating ratio, and the figure-of-merit of the TEG on the power output of the hybrid device were analysed [40]. Different PV cells (c-Si, p-Si, polymer, and CIGS) were also investigated at low and medium concentrations (<50 suns) [31].

Active cooling mechanisms are determinant for hybrid CPV-TEG systems to be feasible at high concentrations [41,33,42,43]. A novel CPV-TEG system integrated with phase-change materials (CPV-PCM-TEG) was proposed in [41]. An optimisation of the structure parameters for maximum efficiency was proposed under different values up to 1,000 × geometrical concentration. The maximum efficiency of the hybrid system with III–V solar cells was 38.9% at 300 K under an optical concentration of 500 suns. The energetic and economic feasibility of hybrid CPV-TEG modules has been recently investigated with an emphasis on actively cooled designs in [33]. The authors theoretically calculated a 43% efficiency at 1,100 suns for the hybrid system compared to a maximum of 40% at 300 suns for the CPV-only system. The CPV-TEG at 1,100 suns resulted in a cost reduction of 71% with respect to the CPV-only system at 300 suns. Thus, increasing the concentration ratio was beneficial for hybridisation in terms of efficiency and cost. Yin *et al.* proposed a novel design method for a CPV-TEG system at a high optical concentration ratio (500 ×). Their model calculates the minimum figure-of-merit of the TEG, thereby enabling the efficiency of the CPV-TEG system to be larger than that of the CPV-only system [42]. Mahmoudinezhad *et al.* evaluated the power generation and efficiency of a CPV-TEG hybrid system under transient solar radiation at 400 suns using a one-dimensional heat transfer model using the energy conservation law. The cooling condition on the cold side of the TEG was assumed to be free convection with water as the working fluid, with a heat transfer coefficient for the heat sink of 2,000 W/(m^2 K). The results show that the efficiencies of the TEG and CPV exhibit opposite trends in response to variations in solar irradiation [43].

Other authors have studied the feasibility of a high CPV-TEG system with passive cooling [44,45]. An experimental and numerical study was conducted by Beeri *et al.* The hybrid system based on MJ solar cells was measured under natural sunlight and concentrated up to 300 suns. The maximum overall efficiency reached 32%, with almost 20% of the direct electrical contribution of the TEG to the hybrid system's efficiency at 300 suns [44]. An electric/thermal/economic model of CPV-TEG modules was recently presented by the authors [45]. We analysed the limits of passive cooling through different thermal resistances of the heat sinks within the range of 800 × –1,900 × geometric concentration. A maximum efficiency of 39.2 % for the hybrid system was achieved at 800 × concentration factor by using low-thermal-resistance heat sinks, while a maximum cost reduction of 46.0% was achieved at the maximum analysed concentration factor (1,900 ×) by using heat sinks with moderate thermal resistance, which was used to predict the annual energy harvesting of different designs of hybrid systems, and an increase in the annual yield of 4.30% was found [46].

Most of the previously published studies were based on a one-dimensional heat transfer analysis. Currently, there are very few studies on three-dimensional (3D) investigations using the finite-element method (FEM) to design and optimise hybrid CPV-TEG systems [47,48,49,50]. The performance of a hybrid CPV-TEG with perovskite solar was studied by introducing a 3D numerical model in [47]. The addition of a thermoelectric material added 0.8% to the overall generation efficiency. Less complex 3D numerical models for a PV module and a TEG device were developed in [48]. Our proposed numerical model ignores the inner structural complexities and considers the device as a homogeneous medium. Simulations were performed for different numbers of TEG modules and low optical concentration ratios (4–9 suns). A comparative study of a CPV-TEG system with and without a flat-plate heat pipe was performed in [49]. They considered temperature-dependent thermoelectric material properties to increase simulation accuracy. The results demonstrated that the overall efficiency was 1.47% and 61.01% higher than that of the PV-TEG and PV-only systems, respectively, at a concentration ratio of 6 × . Shortly after, Shittu *et al.* carried out a 3D numerical investigation of the effect of contact resistance on the performance of CPV-TEG [50]. They observed that the thermal contact resistance between the TEG and heat sink, and that between the PV-TE interfaces was the most important contact resistance, which should be reduced. All authors agree that the 3D numerical models provide a useful tool for investigating and designing hybrid systems owing to the implications of the geometric factors on the system performance. Despite this, in the reviewed literature, there are no 3D studies that consider hybrid systems at UH concentration ratios (>2,000 ×).

We herein present an accurate 3D numerical model by FEM, also

validated with previous work available in the literature, to investigate the potential of a novel hybrid CPV-TEG receiver at UH levels of concentration up to 4,000 × . With this model, the influence of the key design parameters (such as solar cell area, temperature coefficient of efficiency, area ratio, thermoelectric leg height, fill factor, *ZT* figure-of-merit, or heat-exchanger thermal resistance) on the thermal and electrical performance of the system is analysed in detail. Efficiency metrics are introduced to compare the hybrid system with the CPV-only system and evaluate the efficiency improvement achievable with hybridisation. Subsequently, an optimum future design of hybrid CPV-TEG at 4,000 suns is proposed, assuming that the progress in concentrator solar cell development, optics, and thermal management will overcome the limitations of UHCPV in the next few years. The efficiency of this future receiver is discussed to demonstrate the potential of the hybrid CPV-TEG technology at the UH concentration. Our results herein will help in the design of future prototypes that take advantage of both ultra-high concentration levels and hybridisation with thermoelectric generators.

2. Modelling outline

2.1. CPV-TEG receiver structure

A passively cooled CPV-TEG receiver inspired by a previously proposed design [45] was considered for the numerical simulations. The structure of the receiver is shown in Fig. 1. This geometry comprises a standard triple-junction (3 J) GaInP/GaInAs/Ge solar cell mounted on a copper heat spreader through a lead-free SnAgCu solder. The 3 J solar cell was thermally modelled as a single block of Ge [26,51,52] because the top and middle subcells are much thinner than the bottom subcell and, therefore, they can be neglected in the thermal model. A copper plate was used to homogenise and spread out the heat to the hot side of the TE module and accelerate the heat transfer between the solar cell and the hot side. The thermal spreader area is larger than the solar cell area, and is the same as the absorbed area of the TE module. In addition, the copper plate was thermally insulated on the upper surface to obtain the maximum possible excess heat through the TE module. The TE module is coupled between the copper heat spreader and the heat exchanger using a standard adhesive thermal interface material (TIM) on both sides. The TE module converts part of the thermal energy removed from the rear side of the cell to electric power. Most thermal energy is removed from the cold side. Finally, the cold side was cooled by a passive heat exchanger. The physical properties of the different material layers are listed in Table 1. Electrical conductivity has been set only for materials whose constitutive equations for current conservation are solved.

The electrical conductivity of the metallic interconnectors ($\sigma_{Cu}(T)$) has been defined as a linear temperature-dependent function with a reference resistivity of 1.63×10^{-8} Ω·m and a temperature coefficient of 0.0040 1/°C for copper at 20 °C [53]. As the thermal and electric properties of TE materials play a vital role in the TEG module performance [54], the semiconductor thermal and electrical conductivities have been defined by polynomial functions (see Table 2) from real data of a commercial TEG module GM200-49–45-25 [55]. These physical parameters are intrinsic properties that determine the quality of the TE material measured through the dimensionless figure-of-merit (*ZT*):

$$ZT = \frac{\sigma S^2}{k} T_{TE} \tag{1}$$

where σ is the electrical conductivity, S is the Seebeck coefficient, k is the thermal conductivity, and T_{TE} denotes the average temperature of the TE material. The efficiency of a thermoelectric power generator or refrigerator is mainly determined by the *ZT* value and the Carnot efficiency. To analyse the potential of CPV-TEG for a range of TE materials, two *ZT* values were considered in this study: $ZT = 1$ (commercial module GM200-49–45-25) and $ZT = 2.4$ (thin-film p-type Bi_2Te_3/Sb_2Te_3 super lattice [37]). In the case of $ZT = 2.4$, to maintain the temperature dependency in the simulations for the intrinsic properties of the TE material, a *ZT* curve with maximum values around 2.4 was modelled from extracted data from the literature. This *ZT* characteristic was obtained by using the electrical and thermal conductivities of the commercial module and modifying the polynomial coefficients of the Seebeck parameter of the commercial module to fit the literature data. The numerical values of the polynomial functions that represent the thermal conductivity (k_p, k_n), electrical conductivity (σ_p, σ_n), Seebeck coefficient for $ZT = 1$ (S_p, S_n), and Seebeck coefficient for $ZT = 2.4$ ($\acute{S}_p$, $\acute{S}_n$) are shown in Table 2. The *ZT* characteristic curves as a function of the temperature generated from these polynomials are plotted in Fig. 2.a for the case of $ZT = 1$ and $ZT = 2.4$ and compared it to the *ZT* values reported in [37]. The regression analysis between the literature and modelled values for $ZT = 2.4$ is shown in Fig. 2.b. A good linear correlation of $R^2 = 0.9881$ with the literature values was obtained by adjusting the polynomial functions of the Seebeck coefficients. Fig. 2a shows how the value of ZT = 1 is provided for an average TEG temperature of 350 K, while ZT = 2.4 is obtained at 320 K. Currently, there is no commercial TE device with ZT > 2 that can be integrated in a CPV receiver. The purpose of modelling a TEG with ZT = 2.4 is to achieve an approximate simulation of a possible future device that would help in improving the efficiency of ultra-high concentrator systems that allows

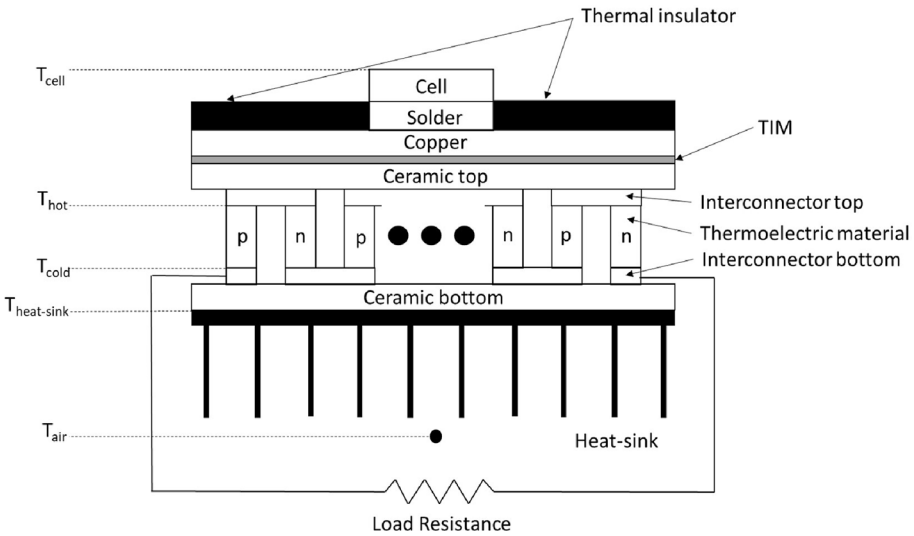


Fig. 1. Structure of passively cooled CPV-TEG receiver based on the design introduced in Ref. [45]

Table 1
Physical properties and layer structure of the materials employed in the hybrid system.

Layer	Material	ThermalConductivity(W/mK)	ElectricalConductivity(S/m)	Thickness(mm)
Solar Cell	Ge	60	–	0.18
Solder	SnAgCu	78	–	0.125
Spreader	Cu	400	–	0.3
TIM	Epo-tek	2.83	–	0.125
Ceramic	AlNi	285	–	0.2
Solder	SnAgCu	78	–	0.125
Interconector	Cu	400	$\sigma_{Cu}(T)$	0.1
TE leg N-type	Bi_2Te_3-N	$k_n(T)$	$\sigma_n(T)$	0.4–2.0
TE leg P-type	Bi_2Te_3-P	$k_p(T)$	$\sigma_p(T)$	0.4–2.0

Table 2
Polynomial functions of TE material properties taken from the commercial module GM200-49–45-25 [55] and, modelled Seebeck coefficients for $ZT = 2.4$ ($\acute{S}_p$, $\acute{S}_n$).

Bi_2Te_3 type	Parameter	Polynomial coefficients $a(T^2)$	$b(T)$	$c(offset)$	Units
p-type	k_p	$0.00361558 \times 10^{-2}$	$-2.6351342 \times 10^{-2}$	622.162×10^{-2}	W/mK
	σ_p	$0.015601732 \times 10^{2}$	-15.708052×10^{2}	4466.38095×10^{2}	S/m
	S_p	$-0.003638095 \times 10^{-6}$	$2.74380952 \times 10^{-6}$	$-296.214286 \times 10^{-6}$	V/K
	S'_p	$-0.007094285 \times 10^{-6}$	$4.66447618 \times 10^{-6}$	$-399.889286 \times 10^{-6}$	V/K
n-type	k_n	$0.00334545 \times 10^{-2}$	$-2.3350303 \times 10^{-2}$	560.6333×10^{-2}	W/mK
	σ_n	0.01057143×10^{2}	-10.16048×10^{2}	3113.71429×10^{2}	S/m
	S_n	$0.001530736 \times 10^{-6}$	$-1.08058874 \times 10^{-6}$	$-28.338095 \times 10^{-6}$	V/K
	S'_n	$0.00286248 \times 10^{-6}$	$-1.67491255 \times 10^{-6}$	$-80.1968088 \times 10^{-6}$	V/K

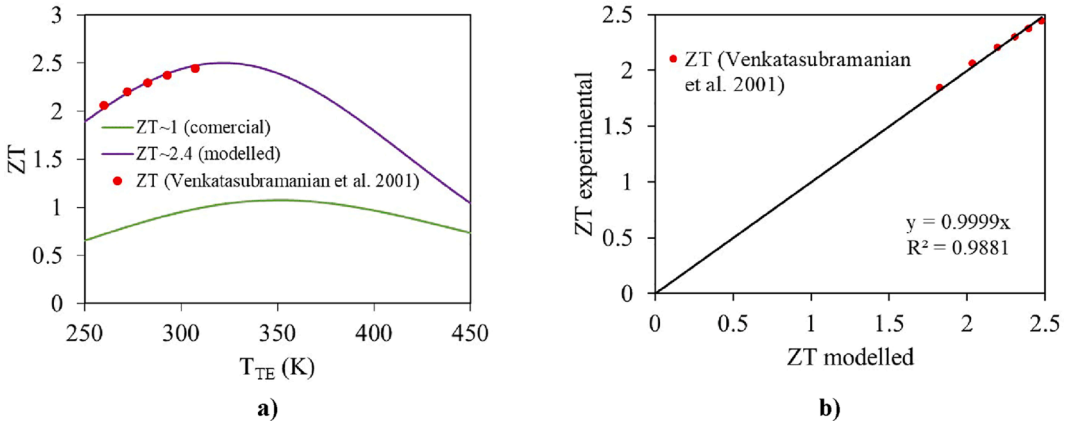


Fig. 2. a) *ZT* characteristic curves as a function of temperature generated from the polynomials in Table 2 for $ZT = 1$ and $ZT = 2.4$, and *ZT* values reported in Ref. [37]; b) Linear regression of the points extracted from the literature and modelled values of ZT = 2.4 by adjusting the polynomial coefficients of the Seebeck parameter.

the potential of such systems to be analysed. For this purpose, the thin-film super lattice from Venkatasubramanian *et al.* [37] was chosen as a reference for modelling.

2.2. Mathematical modelling

Simulations of the thermal and electric problems were performed using the commercial software COMSOL Multiphysics®, which is based on finite-element analysis (FEA). The numerical configuration and mesh properties are described in the following subsections. A heat transfer model for solids combined with an electric current model was selected for modelling the Peltier–Seebeck–Thomson effects. The concentrated photovoltaic, thermoelectric generator, and hybrid system models used herein are described below, along with the boundary conditions set for the simulations.

2.2.1. Formulation for CPV model

The temperature distribution in the CPV system is governed by the energy conservation equation for the steady state:

$$-\nabla\hat{A}\cdot([\boldsymbol{k}]\hat{A}\cdot\nabla\boldsymbol{T}) = \dot{\boldsymbol{q}} \tag{2}$$

where $[k]$ is the tensor of the thermal conductivity, T is the scalar temperature field, and $\dot{q}$ is the volumetric heat source. The source term corresponds to the wasted heat rate produced in the solar cell, as defined by the difference in volume between the total incident irradiance on the receiver ($Q_{rec} = DNIA_{cell}C_g\eta_{opt}$) and the electric power output in the solar cell ($P_{CPV} = DNIA_{cell}C_g\eta_{opt}\eta_{cell}$), resulting in the following equation:

$$\dot{\boldsymbol{q}} = \frac{\boldsymbol{DNIA_{cell}C_g\eta_{opt}(1-\eta_{cell})}}{\boldsymbol{V_{cell}}} \tag{3}$$

where V_{cell} is the solar cell volume, *DNI* is the direct normal irradiance, A_{cell} is the solar cell area, and C_g and η_{opt} denote the geometric

concentration and optical efficiency of the concentrator, respectively. The solar cell efficiency, η_{cell}, is modelled by assuming the impact of temperature rise as follows:

$$\eta_{cell} = \eta_{ref,cell}[1 + TC(T_{cell} - 298.15[K])] \quad (4)$$

where $\eta_{ref,cell}$ is the reference efficiency, TC is the temperature coefficient of efficiency, and T_{cell} is the average temperature of the solar cell in Kelvin. Hence, the efficiency of the CPV system is expressed as follows:

$$\eta_{CPV} = \frac{P_{CPV}}{DNIA_{cell}C_g} = \eta_{opt}\eta_{cell} \quad (5)$$

An important geometric parameter of the hybrid receiver is the spreading area ratio (AR), which is defined as the ratio of the TEG area (A_{TEG}) to the solar cell area (A_{cell}):

$$AR = \frac{A_{TEG}}{A_{cell}} \quad (6)$$

2.2.2. Formulation for TEG model

In a thermoelectric analysis for steady-state conditions, the equations of heat flow ($\vec{q}$) and continuity of electric charge ($\vec{J}$) are coupled by a set of thermoelectric constitutive equations [34]:

$$\vec{q} = T\hat{A}\cdot[S]\hat{A}\cdot\vec{J} - [k]\hat{A}\cdot\nabla T \quad (7)$$

$$\vec{J} = [\sigma]\hat{A}\cdot\left(\vec{E} - [S]\hat{A}\cdot\nabla T\right) \quad (8)$$

where $[S]$, $[\sigma]$ and $[k]$ are the Seebeck coefficient, electrical conductivity, and thermal conductivity, respectively. The electric field $\vec{E}$ can be derived from the electric scalar potential φ in the absence of time-varying magnetic fields, as follows:

$$\vec{E} = -\nabla\varphi \quad (9)$$

Substituting equations (7), (8) and (9) on simplified equations for current ($\nabla\hat{A}\cdot\vec{J} = 0$) and energy ($-\nabla\hat{A}\cdot\vec{q} = \dot{q}$) conservation, the governing equations solved by COMSOL Multiphysics® for temperature distribution and electric potential in steady-state condition becomes

$$-\nabla\hat{A}\cdot\left(T\hat{A}\cdot[S]\hat{A}\cdot\vec{J}\right) - \nabla\hat{A}\cdot([k]\hat{A}\cdot\nabla T) = \dot{q} \quad (10)$$

$$\nabla\hat{A}\cdot([\sigma]\hat{A}\cdot[S]\hat{A}\cdot\nabla T) + \nabla\hat{A}\cdot([\sigma]\hat{A}\cdot\nabla\varphi) = 0 \quad (11)$$

The heat source term $\dot{q}$ shown in the equation above includes the electric power ($\vec{J}\hat{A}\cdot\vec{E}$) spent on Joule heating and on work against the Seebeck field ($[S]\hat{A}\cdot\nabla T$), where the temperature dependence of the Seebeck coefficient has been considered to emphasise the Thomson effect. Once the electric potential distribution is obtained, the thermoelectric generator electrical performance can be derived from the following expressions:

$$P_{TEG} = \frac{V_{TEG}^2}{R_L} = V_{TEG}\hat{A}\cdot I_{TEG} \quad (12)$$

$$\eta_{TEG} = \frac{P_{TEG}}{Q_{hot}} \quad (13)$$

where V_{TEG} is the potential difference in the external load resistance terminals, I_{TEG} is the current intensity, R_L is the matching load resistance, and Q_{hot} is the input heat rate on the TEG module. The current intensity was estimated by integrating the electric charge magnitude ($\vec{J}$) over the surface of the TEG terminals, and the heat rate is obtained by integrating the heat flow ($\vec{q}$) over the hot surface of the TEG module[56]. Another way to evaluate the performance of the TEG module is through TEG effectiveness, which is defined as the ratio between the maximum theoretical efficiency [34] of the TEG module, and the simulated TEG efficiency (Equation (13)):

$$TEG_{Effectiveness} = \frac{\eta_{TEG}}{\eta_{TEG,max}} \quad (14)$$

$$\eta_{TEG,max} = \frac{T_{hot} - T_{cold}}{T_{hot}}\hat{A}\cdot\frac{\sqrt{1+ZT}-1}{\sqrt{1+ZT}+T_{cold}/T_{hot}} \quad (15)$$

where ZT is the dimensionless figure-of-merit, T_h and T_c are the average temperatures in the hot and cold sides of the TE module in Kelvin. The TEG effectiveness metric is introduced in this study to quantify the impact of the internal design of the TEG (TEG area, fill factor, and leg height) on the TE efficiency. The higher the TEG effectiveness, the closer is the TE efficiency to the theoretical efficiency. The theoretical efficiency defined in Eq. (15) assumes that the TE material properties (Seebeck coefficient, resistivity, and conductivity) are not temperature-dependent, the n-type and p-type legs are exactly matched, and 1D heat flow with no other losses is verified.

The cooling mechanism was modelled through its absolute thermal resistance (R_{cool}), which includes the effects of thermal contact resistance, thermal conduction resistance, thermal convective resistance, and thermal radiation resistance, from the cold side of the TE module to ambient temperature. Thus, the surface heat transfer coefficient on the cold side of the TEG module can be described as

$$h_{cool} = \frac{1}{A_{TEG}R_{cool}} \quad (16)$$

2.2.3. Hybrid system efficiency improvement metrics

To evaluate the overall power output of the hybrid system, the photovoltaic and thermoelectric individual power outputs are summed as follows:

$$P_{CPV-TEG} = P_{TEG} + P_{CPV} \quad (17)$$

However, the efficiency of the hybrid system cannot be the sum of the CPV efficiency and TEG efficiency because the individual power input in the solar cell and TE module is different. Therefore, the overall efficiency of the hybrid system was calculated using the following definition:

$$\eta_{CPV-TEG} = \frac{P_{CPV-TEG}}{DNIA_{cell}C_g} = \eta_{opt}[\eta_{cell} + \eta_{TEG}(1 - \eta_{cell})] \quad (18)$$

A comparison of the hybrid CPV-TEG system to the same CPV system operating alone, assuming that the solar cell in the CPV-only system operates at the same temperature as the solar cell in the CPV-TEG system, can be quantified by introducing "temperature-matched relative efficiency improvement" (REI_{TM}) as

$$REI_{TM} = \left(\frac{\eta_{CPV-TEG}}{\eta_{CPV,TM}} - 1\right)\hat{A}\cdot 100 \quad (19)$$

where $\eta_{CPV,TM}$ is the efficiency of the CPV-only system operating at the same temperature as that of the CPV-TEG solar cell. This percentage is always ≥ 0, and it expresses the efficiency improvement in relative magnitude when adding a TEG to the CPV receiver, assuming a temperature-matched condition between both systems.

Although this metric is an indicator of the usefulness of incorporating a TEG (the higher the REI_{TM}, the better the performance of the hybrid system), it does not penalise the possible rise of the cell temperature when adding a TEG to the structure owing to the increase in the thermal resistance of the solar receiver. Because of this, another metric is introduced—"temperature-mismatched relative efficiency improvement" (REI_{TMM}). This metric compares the CPV-TEG system to a typical CPV-only system that uses the same solar cell at the same concentration but is cooled with an aluminium flat-plate heat exchanger. Hence, the cell temperature of the reference CPV-only system was lower than that of the CPV-TEG. The efficiency of this reference CPV-only system was numerically calculated in a previous study [26]. In this previous study,

the same 3 J solar cell properties and the same conditions of DNI and optical concentration as in the current study were analysed. The boundary conditions for the convection and radiation phenomena were also the same. However, an aluminium flat-plate heat exchanger was used to cool the receiver. Consequently, REI_{TMM} is defined as

$$REI_{TMM} = \left(\frac{\eta_{CPV-TEG}}{\eta_{CPV,TMM}} - 1\right) \cdot 100 \quad (20)$$

where $\eta_{CPV,TMM}$ are the efficiencies of the CPV-only system operating at a lower temperature than the CPV-TEG solar cell. This percentage can take values ≥ 0 if incorporating a TEG results in a hybrid system more efficient than a typical CPV-only system with the same solar cell and concentration, or values < 0 if incorporating a TEG is energetically worse than using the CPV alone, owing to the increase in the cell temperature.

2.2.4. Assumptions and boundary conditions

With the aim of simplifying the numerical model, some assumptions have been considered:

- Steady-state conditions were considered in all simulations.
- The 3 J solar cell was modelled as a uniform volumetric heat source assuming that the surface of the cell was homogeneously illuminated.
- The solder layer and TIM layer were modelled as thermal contact resistances, providing the thermal conductivity and layer thickness (see Table 1).
- The TE electrical contact resistance was applied at the interface between the TE materials and copper interconnectors and modelled as a surface impedance (see Table 4).
- The TE thermal contact resistance was applied at the interface between the top and bottom electrical conductors and the top and bottom ceramic plates. An equivalent thin resistive layer contact model was used, and the thermal contact resistance was provided (see Table 4).
- Thermal losses considered in the TEG were selected according to a thermodynamic analysis performed on the internal cavity of the TEG. After a dimensional analysis of thermal convection inside the TEG module, both radiation and conduction through the air gap were considered as the main heat transfer mechanisms to the surrounding air. A low Rayleigh number (Ra) of 4.9060 was obtained, with a limit $Ra < 1700$, from which the natural convective effects are negligible versus the diffusive terms in the air cavity. This result is in agreement with the previous analysis in the literature [57,58].

The following boundary conditions were selected:

- The initial temperature for the solid domain was set equal to the surrounding air temperature, at 293.15.
- As initial surroundings conditions, the concentrator standard operating conditions (CSOC) were used according to IEC 62670–3, that is, an ambient temperature of 20 °C, DNI of 900 W/m^2, and AM1.5d spectrum [59].
- The adiabatic condition was defined on the top surface of the copper spreader according to the proposed design (see Fig. 1).
- Natural convection cooling and radiation to ambient mechanisms were selected for solar cell surface.
- The cooling of the CPV-TEG system was modelled by defining a convective boundary condition on the cold side of the TEG module with the heat transfer coefficient (Equation (16)).
- The air domain was modelled inside the TEG with the physical properties of an ideal gas at 330 K [60].
- For heat loss through radiation, only *surface-to-ambient radiation* has been considered as a boundary condition instead of *surface-to-surface radiation* as no significant difference has been observed between them and the computational cost is lower.

2.3. Computational model validation

The proposed model was validated using two methods. First, a mesh convergence test was performed for the reference case (see Section 3.1 for details on the reference case). The number of mesh elements was increased until the cell temperature and the CPV-TEG power output converged until a relative error of 0.03% was achieved with the previous result. Two control variables were observed to ensure convergence of the temperature and electric potential field. As Table 3 shows, a mesh size of 155,189 tetrahedral elements was selected for all simulations, giving a maximum cell temperature of 371.24 K and power output of 592.63 mW for the reference case. For this mesh size, a computational time of 17 min and 20 s was used.

The simulations were performed on a 3.70-GHz CPU and 16.0-GB RAM. The simulations were run using a segregated approach by default to solve the multiphysical problem. The electrical potential distribution is obtained in the first step using the conjugate gradient method (CG); subsequently, the heat transfer equation for temperature distribution is solved by the generalised minimal residual method (GMRES). Both CG and GMRES are iterative solvers that are more appropriate for ill-conditioned problems. In this case, the radiation process was considered, and material properties together with boundary conditions with unknown temperature dependence were set. These considerations indicate that the algorithm must solve a highly nonlinear system of equations. A relative tolerance of 10^{-4} was established as the stop-criterion convergence of the solution in the iterative solver. With regard to the discretisation scheme, a second-order Lagrange element was chosen for spatial discretisation.

Second, a comparison of the numerical results published in previous literature was made using our model under the same conditions presented in [50]. In Fig. 3, a comparison of our simulated results with the literature data for the hybrid efficiency as a function of TE leg height is shown. The results are in good agreement, with a maximum absolute difference of 0.062% for hybrid efficiency, which corresponds to a relative error of 0.6%.

3. Results and discussion

This section uses the detailed 3D model of the hybrid receiver to analyse the system behaviour and its sensitivity to changes in the different design parameters. In Section 3.1, a reference design case at 2,000 × geometric concentration is defined, and its performance is discussed. This case is also used to validate the method used to obtain the electrical parameters under maximum power conditions by investigating the entire energy balance of the system. In Section 3.2, a parametric analysis is performed by varying the design parameters with respect to the reference case and showing the tendencies in the thermal and performance output variables. In Section 3.3, an optimum design at 4,000 × is proposed and analysed to fully explore the potential of the hybrid systems at UH concentrations.

Table 3
Mesh convergence study.

Nº Elements	$T_{cell}(K)$	% Variation	$P_{CPV-TEG}$(mW)	% Variation
9657	370.14	–	590.21	–
12,612	371.06	0.248	591.25	0.129
20,918	371.16	0.027	590.97	0.047
38,338	371.22	0.016	591.71	0.078
68,722	371.23	0.003	592.54	0.140
155,189	371.24	0.003	592.63	0.015
637,417	371.24	0.000	592.62	0.002

Table 4
Value of the design parameters for the reference case and range considered for parametric analysis.

Parameter	Units	Reference Value	Parametric Value
CPV system			
Cell area	mm × mm	1 × 1	0.5 × 0.5, 1 × 1, 3 × 3
Geometric Concentration	suns	2,000	1,000, 2,000, 4,000
Temperature Coefficient	%/K	−0.05	−0.02, −0.05, −0.10
Optical Efficiency	%	80	–
Reference cell efficiency	%	40	–
TEG system			
Area Ratio	–	13	8, 13, 18
Fill Factor	–	0.5	0.3, 0.5, 0.7
Leg height	mm	1.0	0.4, 1.0, 2.0
Figure-of-merit	–	≈1	≈1, ≈2.4
Number of TE Elements	–	36	–
Electric contact Resistance	$\Omega\ m^2$	1×10^{-9}	–
Thermal contact resistance	$(Km^2)/W$	1×10^{-6}	–
Cooling system			
Thermal Resistance	K/W	5	1, 5, 20
Operating conditions			
DNI	W/m^2	900	–
Ambient temperature	K	293.15	–
Heat transfer coefficient	$W/(Km^2)$	5	–

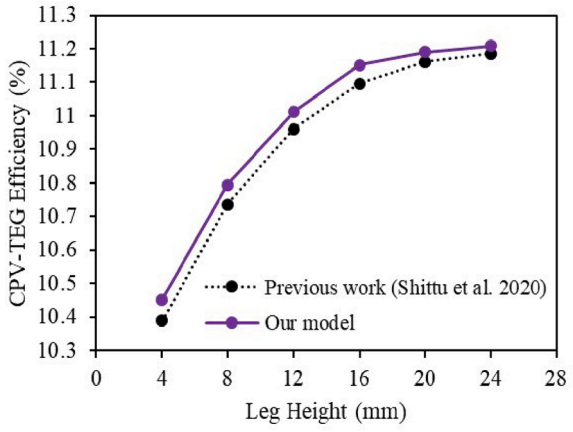


Fig. 3. Overall efficiency of low CPV-TE system as a function of TE leg height. The comparison is between the model presented herein and data in the literature [50] (Cell 1, $R_A = 1$, $A_{TE} = 16$).

3.1. Reference case

Owing to the variety of parameters analysed, a reference case was set with representative values of the design parameters. These values are presented in Table 3. The table also shows the range of values that will be considered in the parametric analysis in Section 3.2. For the reference case, a typical optical efficiency of 80% was considered for the concentrator system [61]. The cell reference value of efficiency is set to 40%, which is in agreement with the tendencies of AzurSpace cell efficiencies for high concentrations [62]. The TE thermal contact resistance and electric contact resistance are 1×10^{-6} $(K\ m^2)/W$ and $1 \times 10^{-9}\ \Omega\ m^2$, respectively, as reported in Ref. [50]. Some key design parameters were allowed to vary within a reasonable range. The geometric concentration was analysed from 1000× (high concentration) to 4000× (UH concentration), with 2000 × as the reference concentration. Small cell areas from 0.5 × 0.5 mm^2 to 3 × 3 mm^2 with different area ratios from 8 to 18 were set according to the tendencies in cell micro-scaling, whereas the reference case uses intermediate values of 1 × 1 mm^2 area and area ratio of 13. The impact of low- and high-temperature coefficients of efficiency was considered with values ranging from − 0.020%/K to − 0.10%/K. The last one represents a typical value for state-of-the-art MJ solar cells [52,62], while lower temperature coefficients are expected to be achievable in the following years (an intermediate value of −0.05%/K was selected for the reference case). The temperature coefficient decreases with concentration [63]; therefore, small coefficients are expected at UH concentrations. Geometrical aspects of TE module, such as leg height and fill factor are varied from 0.4 mm to 2.0 mm and 0.3 to 0.7, respectively (intermediate values of 1.0 mm and 0.5 are set for the reference case). Finally, absolute thermal resistance values of 1–20 K/W were taken to be representative of different cooling systems for a parametric study [19,64,20]. An intermediate value of 5 K/W was chosen as the reference case.

The results of the electrical potential field and temperature field obtained from simulations for the reference case at 2,000 × are shown in Fig. 4. The electrical potential distribution is shown for the entire electrical domain (Fig. 4.A), where the maximum voltage difference provided by the TEG module on external load resistance terminals is 0.34 V. Temperature distributions (Fig. 4.B) provides a maximum temperature of 371 K (98 °C) located in the cell. A maximum temperature gradient of 1 K was applied to the top surface of the solar cell between the corner and middle of the cell. Manufacturers of state-of-the-art MJ solar cells report cell temperature limits around 110 °C [62], whereas current CPV systems are often designed to achieve a 80 °C cell temperature of at CSOC [65]. Although the reference case exceeds the usual operating cell temperature at CSOC in CPV systems, there is empirical evidence that current MJ solar cells can operate at higher temperatures without damage, while showing accelerated degradation over time [66,67]. The obtained maximum cell temperature of 98 °C for the reference case is reasonable considering that incorporating a TEG increases the thermal resistance of the structure and that passive cooling is considered herein. In addition, future solar cells should be manufactured to withstand higher temperatures to take advantage of hybridisation with TEGs. A reasonable level of cell temperature obtained would be possible owing to the micro-scaling of the solar cell (1 × 1 mm^2 in the reference case).

The I–V and P–V characteristic curves of the TEG module for the reference case design are shown in Fig. 5, as obtained from the current and voltage difference at the terminals of the TEG module when the value of R_L is varied (Equation (12)). Therefore, the maximum power point (MPP) of the TEG module was obtained by electric load resistance sweep analysis. Fig. 5 shows that the MPP at reference conditions corresponds to a voltage of 0.34 V and a current of 73.98 mA, thereby providing an electric power output of 25.2 mW. The thermal results of the simulation indicate a temperature difference between the hot and cold sides of the TEG ($T_{hot} - T_{cold}$) of 66.51 °C and a temperature difference between the cold side of the TEG and ambient temperature ($T_{cold} - T_{air}$) of 7.12 °C. The system exhibited the following performance metrics: $TEG_{effectiveness} = 81.68\%$, $REI_{TM} = 4.59\%$, and $REI_{TMM} = 1.63\%$.

An analysis of thermal losses was performed for the reference case. The heat rates of different heat transfer mechanisms are presented in Table 5 to evaluate the heat rate paths given in the 3D model. The total input of the system ($G_0 = DNI\hat{A}{\cdot}A_{lens}$) corresponds to the solar power of the optical system. The highest losses are provided in the optical system, corresponding to 20% of the power input. Focusing on the MJ solar cell, convective losses predominate over radiative losses, but the magnitude of these losses is insignificant in comparison to the power generated. The exceeded heat rate (Q_h) goes from the solar cell to the TEG module through the hot side. Thermal losses in the TEG (conductive + radiative) are notably higher than the losses produced in the solar cell (convective + radiative), as expected. Inside the TEG module, the radiation phenomenon is the main mechanism of heat loss, with similar values to electric power generated, while conductive losses also show an appreciable level. The heat rate extracted by the cooling system (822.37 mW) is still an important amount of the total input power (1,800 mW),

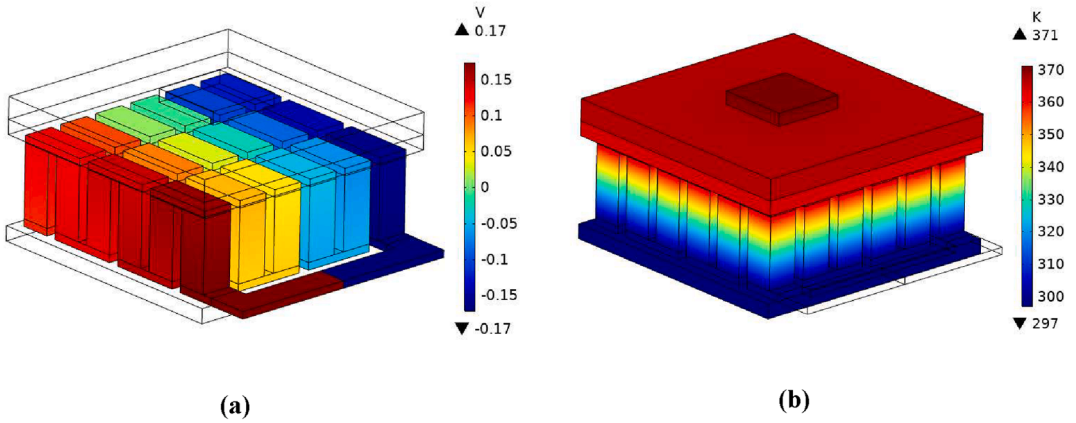


Fig. 4. Result of simulation with reference case parameters of (a) potential field and (b) temperature field.

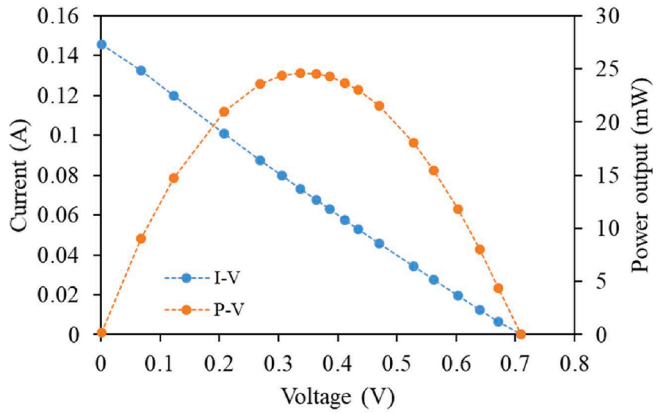


Fig. 5. Electric characterisation of TEG module at reference case configuration.

Table 5
Simulation result of thermal losses analysis in the hybrid CPV-TEG receiver under the reference case (2,000 ×).

Device	*Parameter*	*Heat rate (mW)*	*Percentage (%)*
	G_0	1800.00	100
MJ Cell	Optical Losses	−360.00	20
	Radiative Losses	−0.95	0.05
	Convective Losses	−4.59	0.25
	Electricity Output	−555.00	30.83
	Q_h	879.71	48.87
TEG	Radiative Losses	−19.72	1.096
	Conductive Losses	−12.04	0.67
	Electricity Output	−25.19	1.4
	Q_c	822.37	45.68

indicating that there is much potential for thermoelectric hybridisation to recover the excess heat.

Once the heat rate distribution is obtained, the results of the power output derived from Equation (12) can be tested by applying an energy balance. Considering the TEG module as a closed system, the electric power extracted from the TEG can be obtained in another manner by applying the first law of thermodynamics:

$$P_{TEG} = Q_h - Q_{rad,TEG} - Q_{cond,TEG} - Q_c \tag{21}$$

where $Q_{rad,TEG}$ and $Q_{cond,TEG}$ are the radiative and conductive losses in the TEG, respectively. In this case, an electrical power of 25.58 mW is obtained. This value presents a relative error of 1.52% compared to the value obtained from Equation (12). This error value can be considered acceptable, and the method for obtaining the electrical parameters under the maximum power conditions of the TEG is thus validated.

3.2. Parametric analysis

The parametric analysis was performed by varying several design parameters in the range specified in Table 4. In each simulation, one design parameter was varied, while the remaining parameters were kept the same as in the reference case. The percentage impact on CPV-TEG performance for each design parameter was calculated as the quotient between the difference of the maximum and minimum values of the output variable analysed and the value of the output variable under the reference conditions:

$$\%Impact = \frac{(Z_{max} - Z_{min})}{Z_{ref}} \times 100 \tag{22}$$

where Z is the analysed output variable. The output variables considered are the temperature gradients between the solar cell and hot side of the TEG module ($T_{cell} - T_{hot}$), hot and cold side of TEG module ($T_{hot} - T_{cold}$), cold side of the TEG module and the surrounding air ($T_{cold} - T_{air}$), $TEG_{effectiveness}$ (Equation (12)), REI_{TM}, (Equation (16)), and REI_{TMM}, (Equation (17)).

A summary of the results of the parametric analysis, detailed in depth in the next subsections, is shown in Table 6, indicating the percentage impact of each design parameter on each system output. $T_{cell} - T_{hot}$ is mainly affected by geometric concentration (156%) and solar cell area (135%), while the spreading area ratio is the third most important factor (35%). $T_{hot} - T_{cold}$ is highly affected by geometric concentration (151%) and leg height (146%), followed by the TEG fill factor (90%) and spreading area ratio (73%). Regarding $T_{cold} - T_{air}$, the solar cell area has a significant impact (840%), followed by the heat-exchanger absolute thermal resistance (356%) and geometric concentration (149%). When analysing these temperature outputs, the geometric concentration plays a crucial role in increasing the temperature gap inside the TEG, increasing the TEG efficiency. Decreasing the solar cell area is also crucial for easier cooling of the receiver and for lowering the temperature losses. Increasing the spreading area ratio helps to limit the cell temperature. The intrinsic parameters of the TEG (leg height and fill factor) determine the TEG thermal resistance and, therefore, the temperature gap inside the TEG. In addition, the absolute thermal resistance of the heat exchanger is crucial for decreasing the cold-side temperature.

Regarding the performance metrics, the *TEG effectiveness* is influenced primarily by the leg height (16%), while the spreading area ratio and fill factor play a secondary role (7% and 5%, respectively). The REI_{TM} is affected by the leg height (145%) and geometric concentration (136%) of similar magnitude, followed by the fill factor (80%), spreading area ratio (75%), and ZT (43%). These results highlight that a proper design of the TEG (optimising the leg height, fill factor, and TEG

Table 6
Summary of the results of the parametric analysis.

Parameter	*Impact (%)* Tcell-Thot	Thot-Tcold	Tcold-Tair	TEG effectiveness	REI_{TM}	REI_{TMM}
Cell area	135.11	5.82	840.99	0.85	2.59	125.28
Geom. Concentr.	155.76	151.38	148.65	2.61	135.99	20.43
Temp. Coeff.	3.93	3.92	3.83	0.04	14.09	269.05
Spreading Area Ratio	34.97	72.64	2.93	7.44	75.12	39.30
Thermal Resist.	0.56	0.74	356.53	0.11	0.91	55.21
ZT	0.28	14.55	1.13	1.12	43.42	150.55
Leg Height	3.09	145.96	9.23	16.17	144.68	79.21
Fill Factor	1.69	90.42	2.48	4.88	79.81	21.42

area, and maximising the ZT) is essential to achieve high performance. In the case of the REI_{TMM}, the temperature coefficient of efficiency is the predominant parameter (269%), with the *ZT* figure-of-merit (151%) and the solar cell area (125%) in the second and third places, respectively. In conclusion, for hybridisation to achieve high efficiency gains compared to CPV-only, manufacturing micro-solar cells with a very low-temperature coefficient of efficiency and TEGs with high figure-of-merit is essential. In the next subsections, the trends in the six output variables analysed are discussed, and the main influencing design parameters to provide further details of the conclusions mentioned above.

3.2.1. $T_{cell} - T_{hot}$

The temperature difference $T_{cell} - T_{hot}$ can be considered as a thermal loss that should be minimised with an adequate design of the solar receiver. In Fig. 6, this variable is represented as a function of the geometric concentration for various cell sizes and spreading area ratios. Decreasing the cell size to 1×1 mm^2 or 0.5×0.5 mm^2 is crucial to achieve a low-temperature difference, while commercial 3×3 mm^2 solar cells exhibit considerably higher temperature loss. Increasing the area ratio helps to decrease the temperature loss to some extent. Another interesting result is the linear increase in $T_{cell} - T_{hot}$ with the geometric concentration. This is a non-desirable effect of increasing the concentration (higher luminous flux implies higher temperature drops in the layers of the solar receiver). However, by using small solar cells (1×1 mm^2 or below), the slope decreases considerably. In conclusion, micro-scaling is essential for reducing the temperature loss across the material layers.

3.2.2. $T_{hot} - T_{cold}$

Increasing $T_{hot} - T_{cold}$ helps achieve better TEG efficiencies and enhances the global efficiency of the device. In Fig. 7, this variable is plotted against the geometric concentration for various leg heights and spreading area ratios. The temperature difference linearly increases with the concentration factor, which is one of the main advantages of the UH concentration levels. Increasing the leg height increased the thermal resistance of the TEG, achieving a higher temperature gap inside the TEG. However, there is a limit to increasing the leg height, where the TEG effectiveness begins to decrease, as will be discussed in Section 3.2.5. Finally, decreasing the area ratio helps to increase *the* $T_{hot} - T_{cold}$, achieving better TEG efficiencies. The simultaneous increase in geometric concentration and decrease in area ratio also has limitations when the solar cell exceeds its maximum operating temperature. Cells that withstand higher operating temperatures are recommended to be developed to take advantage of hybridisation with TEGs.

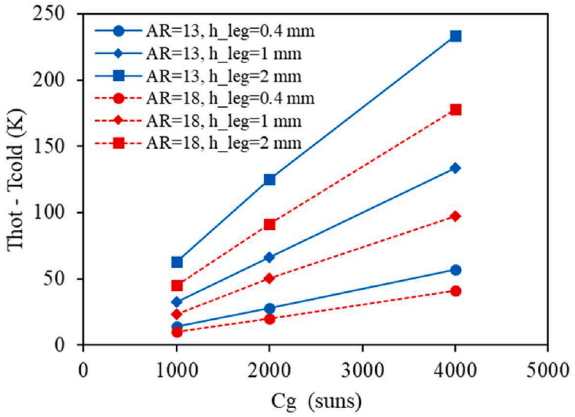


Fig. 7. Temperature difference between the hot and cold sides of TEG as a function of geometric concentration, for different length of TE legs at AR = 13 (blue) and AR = 18 (red).

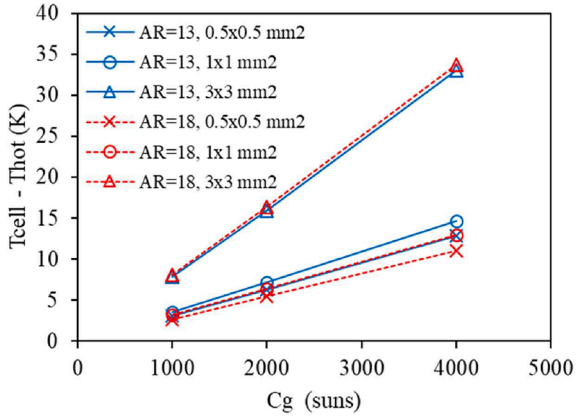


Fig. 6. Temperature difference between solar cell and hot side of TEG as a function of geometric concentration for different solar cell sizes at AR = 13 (blue) and AR = 18 (red).

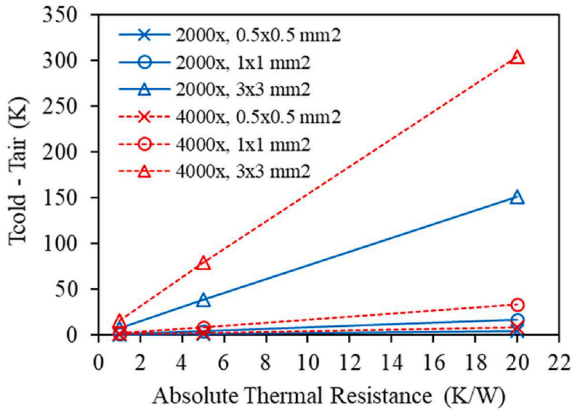


Fig. 8. Temperature difference between cold side of TEG and air as a function of the heat-exchanger thermal resistance for different solar cell sizes at 2,000× (blue) and 4,000× (red) geometric concentration.

3.2.3. $T_{cold} - T_{air}$

The $T_{cold} - T_{air}$ temperature difference needs to be minimised by implementing adequate cooling systems adapted to the cell size. In Fig. 8, the trends of this temperature difference as a function of the heat-exchanger absolute thermal resistance for various cell sizes and geometric concentrations are plotted. The temperature difference increased linearly with the heat-exchanger thermal resistance, and the slope was especially high under UH concentration levels. However, the slope decreased considerably with the micro-scaling of the solar cells. Therefore, the simultaneous improvement of the cooling system and micro-scaling of the receiver are essential factors for achieving excellent performance under UH concentration levels. Under a conventional heat-exchanger thermal resistance of 5 K/W with passive cooling, micro-scaling to 1 × 1 mm^2 cell size or less is required to avoid an excessive increase in T_{cold}, which would imply exceeding the operating cell temperature limits.

3.2.4. $TEG_{effectiveness}$

The $TEG_{effectiveness}$ percentage should be maintained at high values to take advantage of TEG module power generation. A proper design of the TEG module, adapted to the characteristics of the CPV-TEG receiver, is essential to achieve high $TEG_{effectiveness}$; this can be seen in Fig. 9, where this variable is plotted versus leg height for various area ratios and fill factors. Given a fill factor and area ratio value, $TEG_{effectiveness}$ increases with leg height until reaching a maximum and then decreases. Therefore, there is a maximum leg height from which the $TEG_{effectiveness}$ begins to drop for given values of the fill factor and area ratio. For a high area ratio ($AR = 18$), the optimum leg height is outside the analysed range (>2 mm), but this high area ratio would decrease $T_{hot} - T_{cold}$, as shown in Section 3.2.2, indicating that simultaneously increasing the area ratio and the leg height is not a good design option. For more reasonable values of area ratio, the optimum leg height is between 1 and 2 mm (except for a high fill factor value of 0.7, and area ratio of 13, wherein the optimum leg height is again>2 mm). These results highlight the importance of obtaining a good combination of leg height (values in the range from 1 to 2 mm can be suggested), area ratio, and fill factor of the TEG module to obtain $TEG_{effectiveness}$ values higher than 80%.

3.2.5. REI_{TM}

Increasing the REI_{TM} percentage indicates better exploitation of hybridisation. The trends in REI_{TM} as a function of the geometric concentration for several area ratios are shown in Fig. 10. This graph highlights the decisive influence of increasing the geometric concentration and decreasing the area ratio for increasing the temperature gap inside the TEG and obtaining high power gains with the hybridisation.

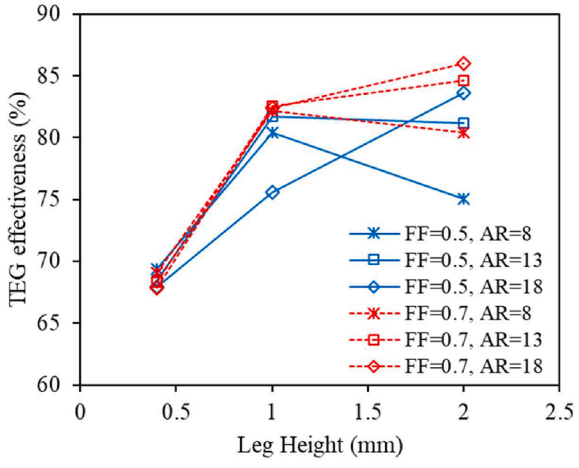


Fig. 9. TEG effectiveness for different spreading area ratios as a function of TE leg height, considering a fill factor of TEG module of 0.5 (blue) and 0.7 (red).

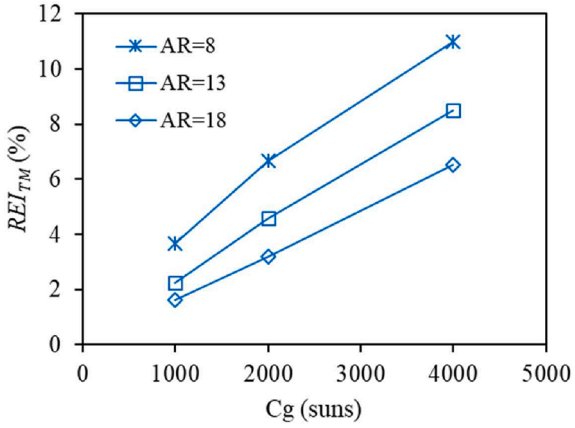


Fig. 10. Temperature-matched relative efficiency improvement as a function of geometric concentration at different spreading area ratios.

At 2000×, a maximum REI_{TM} of approximately 7% was observed, while at 4000×, a maximum of approximately 11% was obtained, considering $ZT = 1$ (reference value). There are also other influencing factors, as shown in Fig. 11, where REI_{TM} is plotted versus leg height, fill factor, and ZT. Increasing the leg height and decreasing the fill factor of the TEG module helped to increase the temperature gap inside the TEG, allowing for better relative efficiencies. The use of TE materials with high $ZT = 2.4$ significantly improves the REI_{TM}, achieving approximately 10% at 2000 × and approximately 15% at 4000 × .

3.2.6. REI_{TMM}

The REI_{TMM} percentage allows for the benefits of the hybrid system compared to a CPV-only system to be quantified. The most influential design factors are the temperature coefficient of efficiency, cell size, and ZT figure-of-merit. The REI_{TMM} is plotted versus the temperature coefficient for various cell sizes and ZTs in Fig. 12. *A* REI_{TMM}>0% is required for hybridisation to make sense. This means that the TEG module provides electricity above the solar cell losses caused by TEG integration. Developing solar cells with lower temperature coefficients leads to us obtaining the advantages of hybridisation. Low-temperature coefficients are expected under the UH. Decreasing the cell size to 1 × 1 mm^2 or below is also essential. Finally, the commercial availability of TE materials with a high ZT will be very useful for increasing the device efficiency. Under conventional temperature coefficients of 0.1 %/K, only high figure-of-merit TE materials integrated with microcells of 1 × 1

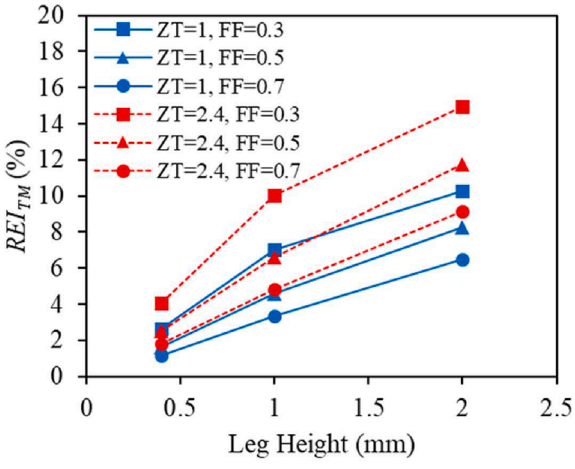


Fig. 11. Temperature-matched relative efficiency improvement as a function of TE leg height, at different TEG fill factors, considering values of ZT = 2.4 (red) and ZT = 1 (blue).

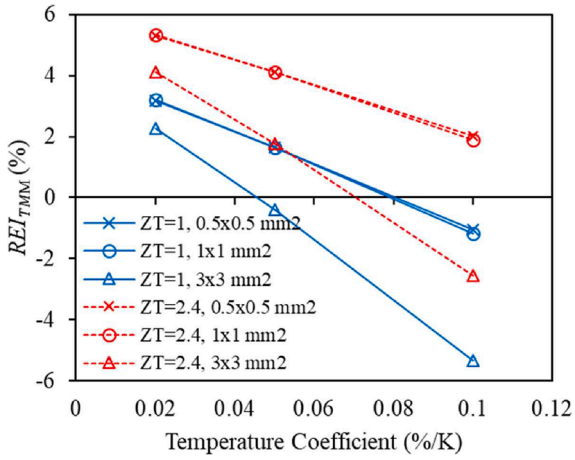


Fig. 12. Temperature-mismatched relative efficiency improvement as a function of solar cell temperature coefficient of efficiency at different cell areas, considering values of ZT = 2.4 (red) and ZT = 1 (blue).

mm^2 or below would be able to take advantage of hybridisation. However, with temperature coefficients lower than 0.08 %/K, standard $ZT = 1$ TE materials can be integrated with microcells to achieve efficiency improvements. REI_{TMM} around 5.5% could be achieved at 2000× (reference case) by using $ZT = 2.4$, and microcells of 1×1 mm^2 or below. REI_{TMM} and global hybrid efficiency are shown in Fig. 13 as a function of geometric concentration. Hybrid efficiency increases as the concentration ratio increases for *ZT* values around 2.4, and the same happens with REI_{TMM}. On the contrary, with a low *ZT,* the performance of the hybrid system starts to decrease above 2.000 suns of geometric concentration. These results show that increasing the concentration is beneficial for hybrid systems if micro-scale solar cells and high-ZT thermoelectric materials are developed and, therefore, justifies the interest of this study.

3.3. Optimum design of UHCPV-TEG receiver at 4000×

In this subsection, the potential of the optimised hybrid CPV-TEG system at the UH concentration ratio is fully explored. Taking into account the results of the parametric analysis, a hybrid receiver was defined at a 4000 × geometric concentration with the design parameters listed in Table 7. A micro-solar cell (0.5×0.5 mm^2) with very low-

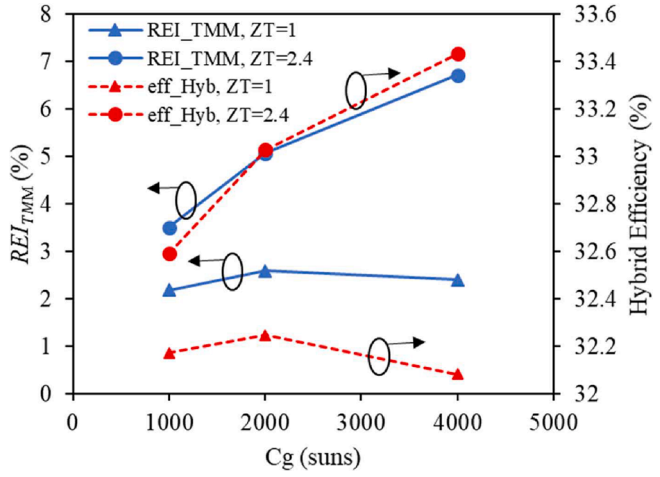


Fig. 13. Temperature-mismatched relative efficiency improvement (blue) and hybrid efficiency (red) as a function of geometric concentration, considering values of ZT = 2.4 (circle) and ZT = 1 (triangle). Conditions: Acell = 1×1 mm^2, AR = 13, TC = − 0.05 %/K and R_{th} = 5 K/W.

Table 7
Design parameters of the optimum UHCPV-TEG receiver for exploring the maximum performance.

Parameter	*Value*	*Unit*
Geometric concentration	4000×	–
Cell area	0.5 × 0.5	mm × mm
Area Ratio	13	–
Fill Factor	0.5	–
Leg height	1	mm
Figure-of-merit	2.4	–
Temperature Coefficient	−0.02	%/K
Thermal Resistance	1	K/W

temperature coefficient of efficiency (−0.02 %/K) has been simulated by using advanced TE materials ($ZT = 2.4$) and very low thermal resistance for the cooling system (1 K/W). The TE module design was adjusted (area ratio, fill factor, and leg height) to not exceed 150 °C cell temperature operating under CSOC (see Table 7). The thermal and performance outputs of the receiver are listed in Table 8. A temperature gap inside the TEG of 110.72 °C would be achievable. This high temperature gap in combination with the high-*ZT* figure-of-merit of the TE materials would allow for important efficiency improvements (REI_{TM} = 11.16%; REI_{TMM} = 10.80%). The cold-side temperature of the TEG remains close to the air temperature ($T_{cold} = 21.87^{\circ}C$)thanks to the micro-scaling of the system, and the use of thermal resistances as low as possible within the limits of the passive cooling mechanism. TEG *effectiveness* was maintained at a high value (99.01%). The results highlight that hybridisation is very attractive at UH. These future performance values could only be achieved with the development of micro-receivers integrated with TE materials with high figure-of-merit and advanced cooling mechanisms. Therefore, promising results should encourage researchers to overcome the barriers that limit the development of such prototypes at UH concentration levels.

4. Conclusions

Concentrator photovoltaic technology will evolve toward UH concentration levels to increase efficiency and lower cost. The operation under UH concentration levels offers an opportunity to take advantage of hybridisation with thermoelectric generators to further enhance efficiency. The use of 3D finite-element models is essential for predicting the performance of such hybrid devices and optimising their design. In this study, a detailed 3D model was used to investigate the thermal and electrical performances of hybrid thermoelectric concentrator photovoltaic receivers at UH concentrations. The influence of the key design parameters was assessed by parametric analysis. The results highlight the usefulness of increasing the concentration factor to achieve better global efficiency. In addition, the micro-scaling of high-temperature solar cells, integration of high figure-of-merit thermoelectric materials, and improvement in cooling systems have demonstrated major impact on the efficiency enhancement of hybrid systems. In the case of the "temperature-mismatched relative efficiency improvement" (REI_{TMM}), wherein the hybrid system is compared with a CPV-alone system, the temperature coefficient of efficiency is the predominant parameter (269%), with the ZT figure-of-merit (151%) and the solar cell area (125%) in the second and third places, respectively. Regarding the thermoelectric generator, obtaining a good geometric design with simultaneous optimisation of the leg height, fill factor, and spreading area ratio is crucial to achieve high thermoelectric effectiveness and

Table 8
Simulation results of the UHCPV-TEG receiver at 4000×.

T_{cell} (°C)	T_{hot} (°C)	T_{cold} (°C)	$TEG_{effectiveness}$ (%)	REI_{TM} (%)	REI_{TMM} (%)	η_{Hybrid}(%)
145.1	132.59	21.87	99.01	11.16	10.80	34.72

maximise profit of the hybridisation.

To explore the potential of the hybrid systems at ultra-high concentrations, an optimum design has been proposed with 0.5 × 0.5 mm^2 solar cells integrated with a thermoelectric generator of $ZT = 2.4$ thermoelectric generator at 4000 × geometric concentration. The temperature coefficient of efficiency and the cooling system thermal resistance decreased to − 0.02 %/K and 1 K/W, respectively. The thermoelectric geometry was adjusted for the cell operating temperature to not exceed 150 °C. The results show that this future receiver achieves a REI_{TMM} of 10.8% compared to a concentrator photovoltaic-only receiver that uses the same solar cell at the same concentration. Therefore, hybridisation shows great potential improvement at UH concentrations. However, for these promising results to be made reality, extensive research is required to cover novel optical systems capable of reaching UH concentration levels with good optical efficiency and acceptance angle, innovative high-efficiency high-temperature micro-solar cell architectures, powerful cooling mechanisms for solar receivers, and improved thermoelectric materials. This study marks the direction in which concentrator photovoltaic technology should align in the next few years and shows the potential benefits of the proposed improved designs.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study is part of the project PID2019-106497RB-I00 funded by the Spanish Ministry of Science and Innovation and FEDER funds, and of the project P18-RT-1595 funded by the Junta de Andalucía [PAIDI 2020] and FEDER funds. P. M. Rodrigo and E. F. Fernández thank the Spanish Ministry of Science, Innovation and Universities for the 2020-2024 Beatriz Galindo [BEAGAL18/00164] and Ramón y Cajal 2017 fellowships [RYC-2017-21910], respectively. A. Valera also thanks the Ayudas predoctorales para la Formación de Personal Investigador con cargo a la Acción 9.a) del Plan de Apoyo a la Investigación de la Universidad de Jaén [RYC_2019_4].

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